

1. Administrative & Study Identification

Full Study Title: An Open-Label, Single-arm Study to Evaluate the Safety and Efficacy of Dabrafenib in Combination With Trametinib in Chinese Patients With BRAF V600E Mutation-Positive Metastatic Non-Small Cell Lung Cancer **Short Title/Acronym:** Dabrafenib + Trametinib in Chinese BRAF V600E NSCLC (CDRB436ECN01) **Protocol Number:** CDRB436ECN01 **NCT Number:** NCT04452877 **Posting ID:** 339605082052121306 **Sponsor/Company Name:** Novartis Pharmaceuticals Corp **Investigational Product:** Dabrafenib (BRAF inhibitor) + Trametinib (Trade Name: Mekinist, ATC Code: L01EE01, MEK inhibitor) **Phase of Development:** Phase 2 **Medical Monitor:** Novartis Clinical Operations Lead / Investigator (e.g., Dr. Li Zhang)

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the overall response rate (ORR) of dabrafenib in combination with trametinib by central independent review per RECIST 1.1 criteria in Chinese patients with BRAF V600E mutation-positive metastatic non-small cell lung cancer (NSCLC). **Secondary Objectives:** To evaluate ORR by investigator assessment, progression-free survival (PFS), duration of response (DOR), overall survival (OS), safety, tolerability, and quality of life (QoL).

2.2 Background & Rationale To address the critical unmet medical need in Chinese patients with metastatic NSCLC harboring BRAF V600E mutations. These patients historically have poorer outcomes and lower response rates to standard first-line chemotherapy compared to other targeted therapies. The combination of dabrafenib and trametinib has shown substantial antitumor activity globally, and this study provides the clinical evidence required to validate the safety, efficacy, and QoL benefits specifically within the Chinese patient population. The general study design was discussed and agreed with China National Medical Products Administration and based on a similar design used in the global pivotal phase II study.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A **Trial Status:** COMPLETED (Study has been completed)

3.2 Planned Enrollment & Duration Target **N**: 40 evaluable Chinese adult patients **Number of Sites**: Multicenter study in China **Estimated Study Start**: 19-Aug-2020 **Estimated Primary Completion**: 28-Dec-2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults with an ECOG performance status ≤ 2 . 2. Histologically or cytologically confirmed diagnosis of Stage IV NSCLC (according to AJCC 8th edition) that is BRAF V600E mutation-positive by local test result from a qualified assay. 3. Measurable disease per RECIST 1.1 and an anticipated life expectancy of ≥ 3 months. 4. Previously treated or untreated for metastatic NSCLC. Previously treated participants must have received ≤ 3 prior systemic therapies (including at least one platinum-based chemotherapy) or objective progression on/after immune checkpoint inhibitor therapy. Participants with EGFR or ALK mutations who have previously received therapy with EGFR or ALK inhibitor(s) are eligible. 5. Adequate bone marrow and organ function (without continuous supportive treatment), and Left ventricular ejection fraction (LVEF) $\geq$ lower limit of institutional normal (LLN).

4.2 Exclusion Criteria 1. Previous treatment with a BRAF inhibitor or a MEK inhibitor. 2. Participants with brain or leptomeningeal metastases that are symptomatic, asymptomatic/untreated > 1 cm, or treated but not clinically and radiographically stable 3 weeks after local therapy. 3. History of another malignancy < 3 years prior to starting study treatment or any malignancy with confirmed activating RAS-mutation. 4. Prior anti-cancer treatment-related toxicities $\geq$ Grade 2 (CTCAE v4.03); prior anti-cancer treatment within the last 2 weeks (or immune checkpoint inhibitors within 4 weeks) preceding the first dose. 5. Cardiac or cardiac repolarization abnormality, history or current risk of retinal vein occlusion (RVO) or serous retinopathy, history or current interstitial lung disease (ILD) or non-infectious pneumonitis. 6. Participants with known HIV, active Hepatitis B infection (HbsAg positive), or a positive test for Hepatitis C ribonucleic acid (HCV RNA). 7. Pregnant or nursing (lactating) women; concurrent participation in other clinical trials using experimental therapies. Sexually active males must use a condom during intercourse for the duration of the study and 16 weeks post-treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Dabrafenib 150 mg twice daily + Trametinib (Mekinist) 2 mg once daily. **Route of Administration**: Oral. **Duration of Treatment**: Continuous until disease progression, start of a new anti-neoplastic therapy,

unacceptable toxicity, pregnancy, withdrawal of consent, lost to follow-up, physician's decision, death, or trial termination.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing disease-related symptoms and toxicities (e.g., antipyretics for pyrexia). **Prohibited:** Concomitant anti-cancer therapies, concurrent use of other investigational agents, and medications known to be strong inhibitors/inducers of CYP3A4 that could interact with the investigational products.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Imaging (CT/MRI/ECHO/MUGA), BRAF V600E testing, Baseline PROs
Baseline	Day 0	First Dose Dispensation, Baseline Safety Labs, Vitals
Interim	Weeks [X] $\pm$ [Y]	Tumor assessments (RECIST 1.1), Safety Labs, AE Monitoring, QLQ-C30 Questionnaires, IP Accountability
End of Study	Variable	End of Treatment Visit, Final Scans, Follow-up for Survival and Progression

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Overall Response Rate (ORR), defined as the proportion of participants with best overall response (BOR) of complete response (CR) or partial response (PR). **Measurement Method:** Central Independent Review Assessed by RECIST v1.1.

7.2 Secondary & Exploratory Endpoints * Overall Response Rate (ORR), Investigator Assessed by RECIST v1.1 * Progression Free Survival (PFS), Investigator Assessed by RECIST v1.1 * Duration of Response (DoR), Investigator Assessed by RECIST v1.1 * Overall Survival (OS) * Safety, Tolerability, and Quality of Life (QoL via EORTC QLQ-C30)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring at each clinical visit. Frequent treatment-related AEs monitored included pyrexia (45%), AST increase (35%), and decreased neutrophil/white blood cell counts. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness to the sponsor. **Data Monitoring Committee (DMC):** Novartis internal safety review committees and central independent radiological review board.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for baseline and efficacy analyses; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** A sample size of ≈ 40 participants was planned and deemed adequate to assess the ORR and safety profile of dabrafenib + trametinib in this specific Chinese mutation-positive cohort. **Handling of Missing Data:** Censoring rules applied to time-to-event endpoints (PFS, DOR, OS) according to standard FDA/EMA oncology guidelines; LOCF or mixed-effect models for QoL questionnaire data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard clinical site monitoring with centralized review of radiological imaging to ensure objective RECIST 1.1 assessments. **Audit Trail:** Validated Electronic Data Capture (EDC) system with full audit trails for eCRF lock and data clarifications.

11. Study Identifiers & Governance

Primary ID: CDRB436ECN01 **Posting ID:** 339605082052121306 **Registry Number:** NCT04452877 **Sponsor/Lead Organization:** Novartis **Study Title:** A Study of Dabrafenib in Combination With Trametinib in Chinese Patients With BRAF V600E Mutant Metastatic NSCLC **Official Regulatory Title:** An Open-Label, Single-arm Study to Evaluate the Safety and Efficacy of Dabrafenib in Combination With Trametinib in Chinese Patients With BRAF V600E Mutation-Positive Metastatic Non-Small Cell Lung Cancer

12. Clinical Framework (Oncology / NSCLC Specific)

Primary Condition / Disease Areas: Carcinoma, Non-Small-Cell Lung **Preferred UMLS Name:** Non-small cell lung carcinoma with BRAF V600E mutation (MeSH Code: D002277 / Concept: Carcinoma) **Diagnosis Threshold:** Genomic confirmation of BRAF V600E mutation (NMPA and/or MOH-approved assay) **Medical Methodology:** Precision Targeted Therapy (BRAF and MEK inhibition) **Optimization Target:** Maximizing tumor response (ORR) and extending progression-free survival while maintaining Quality of Life (QoL)

13. Study Procedures & Methodology

Management Pathway: Standard oncological pathways for oral targeted therapies. **Education Protocol (HCP):** Site initiation

visits covering RECIST 1.1, AE management (specifically pyrexia handling protocols), and dosing interruptions/modifications. **Education Protocol (Patient):** Strict guidance on taking continuous daily oral doses, recognizing symptoms of fever/infection, and completing QoL surveys. **Quality Audit Mechanism:** Central independent radiological review for unbiased response assessment and rigorous drug accountability logging.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up continues until disease progression, death, or trial termination. **Onsite Visit Cadence:** Standard oncology treatment cycles (e.g., Day 1 of each 3-to-4-week cycle) for dispensing, scans every 6-8 weeks initially. **Remote Monitoring Frequency:** Follow-up for overall survival via telephone every 12 weeks post-progression. **Checklist Review Interval:** Patient-reported outcomes (PROs) / EORTC QLQ-C30 administered at specific treatment milestones.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				